

12 December 2022

Prediction of the Stock Market Based on Machine Learning and Sentiment Analysis

Prajwal Jishtu¹, Harshil Prajapati, Jinan Fiaidhi

1. Lakehead University

Abstract

Nothing is rock-steady in the stock market, which is a very volatile market. Nevertheless, there are a variety of ways and approaches one may utilise to learn about this dynamic movement and be prepared for it as technology develops. The focus of this essay is on different methods for quickly identifying market trends. The suggested strategy is comprehensive because it includes pre-processing the stock market dataset, a range of feature engineering techniques, and the integration of a customised deep learning-based system for forecasting stock market price patterns. The best and most suggested method for prediction is the model with the least amount of error. In order to conduct this study, we used three distinct models and ran sentiment analysis on news articles mentioning the firm or the stock. The results of this classification have given investors additional information to help them make decisions about where to stake their money as well as clear and incisive insight into the market's irregular ups and downs.

Keywords

arima, computing and processing, engineering profession, lstm, sarima, sentiment analysis (sa), stock market prediction

Prediction of the Stock Market Based on Machine Learning and Sentiment Analysis

1st Harshil Prajapati
Department of Computer Science
Lakehead University
hprijap3@lakeheadu.ca

2nd Prajwal Jishtu
Department of Computer Science
Lakehead University
pjishtu@lakeheadu.ca

3rd Dr. Jinan Fiaidhi
Department of Computer Science
Lakehead University
jinan.fiaidhi@lakeheadu.ca

Abstract—Nothing is rock-steady in the stock market, which is a very volatile market. Nevertheless, there are a variety of ways and approaches one may utilise to learn about this dynamic movement and be prepared for it as technology develops. The focus of this essay is on different methods for quickly identifying market trends. The suggested strategy is comprehensive because it includes pre-processing the stock market dataset, a range of feature engineering techniques, and the integration of a customised deep learning-based system for forecasting stock market price patterns. The best and most suggested method for prediction is the model with the least amount of error. In order to conduct this study, we used three distinct models and ran sentiment analysis on news articles mentioning the firm or the stock. The results of this classification have given investors additional information to help them make decisions about where to stake their money as well as clear and incisive insight into the market's irregular ups and downs.

Index Terms—Sentiment Analysis, ARIMA, SARIMA, LSTM, Stock Market Prediction.

I. INTRODUCTION

The securities exchange has grown to be a substantial component of an economy, since it enables people to speculate and earn enormous sums of money. The securities exchange has turned into a significant part of a country's economy since it permits individuals to make speculations and gain enormous amounts of cash. A financial exchange is an arrangement of monetary trades where share dealings are conducted. The ownership claims on firms are mirrored in an equity market or share market. Shares from individual transactions or the public stock exchange, such as shares of private corporations sold to investors, may fall under this category. Trading in the stock market is the transfer of funds from speculative investors to large institutional investors, such as banks, firms, etc. However, because unpredictable behavior is seen, stock market investments are viewed as a very risky activity [1].

There has been a lot of research for a very long time in the field of stock market prediction. It becomes more exciting as new technologies are discovered. Despite much research on stock market forecasting, there is still no proven clear mechanism for doing so. Today's main financial organizations and all academic experts are focused on maximizing returns with the least amount of risk. Therefore, it is essential to be aware of the stock market's influencing elements in order to reduce risk. Fundamental, technical, and market sentiment elements are grouped together as the three most significant influences on the stock market. Market sentiment has long

been thought to be an unclear element with significant and biased implications on stock prediction [2]. According to the efficient market hypothesis (EMH), social media emotions can aid investors maximize their returns by promoting a greater understanding of the stock market. As a result, opinion mining, also known as sentiment analysis, which gathers data from sentiments, can help us better comprehend the stock market. Even though stock market forecasting has been a topic of continuing research, quantitative analysts and academics have never found it to be simple. According to the efficient market hypothesis (EMH), it is hard to accurately forecast the market in every aspect. Despite this, researchers and traders continue to try to create trading approaches that can outperform the stock price random walk pattern. Several studies have shown that changes in stock market values are not random and can even be slightly forecast. The answer to improving stock price prediction accuracy is said to be in recognizing stock patterns and outside influences [2].

The efficient market hypothesis states that since stock prices are the outcome of reasonable expectations and information, mood has an impact on financial markets and stock prices. In other words, a portion of the stock price reflects expectations for the company's future performance. Using user opinions to improve stock price prediction has been a popular research area due to the accessibility of enormous corpora of user opinions via social media and the effectiveness of machine learning algorithms for large-scale text mining [3].

News is important in the field of investing because it gives investors information, they need to make judgments on the stock markets. People's emotions and ideas can be shaped and influenced by it, which affects how they decide whether to buy or sell in markets. The collapse of the world's markets in March 2020 during the COVID-19 pandemic serves as a recent example of this. Due to the enactment of extensive lockdowns compelling companies to shut down and cease operations, investors were left in the dark, which resulted in the global markets collapsing in March 2020 [4]. To find significant information, one can examine a variety of textual data sources, such as tweets, annual reports, news articles, and more. The best source of concealed information is textual data, especially news, as compared to numerical data because it enables the forecasting of financial trends using its logic. For instance, a fall in the company's shares can be predicted by an investor if a news story about the company mentions

”resignation” or ”danger of default.” Furthermore, news about a variety of unforeseen concerns may have an impact on stock market trends. Such events include, among others, civil unrest, war, terrorism, political and economic shocks, and natural disasters. Better knowledge finding algorithms from textual data are therefore desperately needed.

II. LITERATURE REVIEW

Sentiment analysis is a technique for classifying opinions based on the language that has been provided, specifically to determine whether or not users are thinking positively, negatively, or neutrally about a certain topic or piece of news. Typically, lexicon-based or machine learning-based methodologies will be used to analyze sentiment.

Major shareprice prediction systems are built on the fundamentals of technical and fundamental analysis, however recent research has revealed that stock prices are strongly correlated with corporate news [5]. Fundamental analysis examines stock trend in relation to economic variables. Technical analysis uses historical quantitative values, such as volume and market value, to forecast the exchange. We tend to think that historical stock trend patterns will eventually repeat themselves when we investigate this example [6].

In [7], Chen & Lazer looked into the relationship between twitter feeds and stock market activity. They employed classification and regression approach to anticipate the changes in the stock market and had a wide range of options that may influence sentiment levels. Zhang has shown a link between stock prices and Twitter sentiment, and the terms used in tweets also show a link to changes in prices. They compared the outcomes of three machine learning methods in addition to how well they affected correlation [8].

The Hidden Markov Model (HMM) was employed by Hassan and Nath in [9] to forecast the stock market values of four significant airlines. The model’s states are divided into four groups using the closing price, opening price, lowest price and maximum price. The methodology used in this research has the advantage of not requiring specialized expertise in order to develop a prediction model. Despite being restricted to the airline business and assessed on a short dataset, this research might not produce a generalizable prediction model. The comparative work can use a technique from stock market forecasting operations.

Some analyses in [10] were exclusively supported by a pattern. Since Deep Learning (DL) algorithms will only acknowledge patterns and correlations in data, it is easier and more popular to make stock market predictions using DL. The accuracy of DL methods is higher despite being used for high frequency stock market prediction [11]. Support Vector Regression (SVR), Random Forest, and K-nearest neighbour are a few alternative machine learning techniques that have been extensively used in prior research. Because time series data or statistic information make up a substantial portion of stock market information, LSTM offers improved accuracy in stock market prediction [12]. Recurrent neural networks (RNNs)

have been modified in order to address the issue of long-term dependency in conventional RNNs. Along with these quantitative methods, it is also common to predict stock market trends by examining text data, such as events, announcements, news, and social opinion. Utilizing this textual data, we will perform a sentiment analysis [13] to determine whether the trend of the stock is rising or decreasing. By using the Twitter dataset for sentiment extraction, the author of [14] efficiently compares the lexicons, demonstrating that SentiWordNet and MPQA (Multi-Perspective Question Answering) perform more effectively than the others. However, because lexicon-based approaches rely on static lists, they are unable to handle new words in an efficient manner. To solve this problem, many works encrypt the emotion score using statistics and machine learning techniques. Restricted analysis has so far been performed on StockTwits data [15].

Feature extraction, which reduces input data into a more manageable format for future processing, is a crucial part of prediction. Most early studies on news-sensitive market trend prediction employed shallow, unstructured feature extraction methods that used words as features. Bag-of-words (BoW), noun phrases, named entities are a few examples [16].

A deep learning system that was trained on a common set of financial market features was employed by Sirignano and Cont in [17]. The dataset used included buy and sell records for all transactions as well as order cancellations for around 1,000 NASDAQ securities obtained from the stock exchange’s order book. The NN consists of three layers of LSTM units, and the stochastic gradient descent (SGD) approach is used to optimize the feed-forward layer with rectified linear units (ReLU) in the final layer. In addition to the stocks in the training set, they were able to generalise and include other equities in their universal model. Despite highlighting the advantages of a universal paradigm, they have expensive training requirements. It is currently unknown if any useless features were contaminated with the data before being fed into the model because of the implicit programming of the deep learning method. The authors found that feature selection may successfully be used to reduce computational complexity and that it would have been preferable to undertake feature selection before training the model.

However, the structured feature extraction method combines verbs, nouns, and words. In contrast to structured feature extraction methods, shallow feature extraction techniques are unable to capture the entirety of an event as structured entity-relation information. It is therefore challenging to illustrate how news events impact stock market trend forecast due to superficial aspects [18].

III. METHODOLOGY

The design of this research is based on both Qualitative and Quantitative as we must collect news and social media updates for the sentiment analysis and even take the historical data and apply different machine learning techniques to predict. The data collection will be as secondary source as we must take historical data which will already be available and for

sentiment analysis, we would need the sentiment on the news article collected by APIs and web scraping too. The prediction of Stock market and sentiment analysis will be done by the algorithms which will be used. Comparing and getting the best results will be done by these algorithms.

A. The Collections of Datasets

The method will be started with the collection of datasets. As we need two datasets for the process. We will download the historic datasets for stock market and for sentiment analysis we will collect the data through API and web scraping. For downloading the stock data we will use Yahoo Finance which will have opening, closing, highest, lowest, volume and the adjusted close price [19]. By using this method we will have the ability to define a string of rules by some selected keywords [20]. As we take an example we can have specific keywords which would define a string by any name as company name with highlighting of terms such as open, close etc.

Investing.com will be used for the news article for sentiment analysis by web scraping. We would be considering Beautiful soap and Selenium for that. Before combining the data for the stock price we have to find out the sentiment score as it will indicate which direction the sentiments new article are influencing. We consider this step at first because we have categories in which articles are classified, Positive which means the stock is high, Negative which means it is low and Neutral which has no flow as the article cannot find it out or in which it is not sure.

B. Pre-Processing of Data

Stop words, tokenization, special characters, URLs will be removed from the data. Cleaning of the data will be performed except punctuation because they are very common. In cleaning of data, word negation, repeated words are removed [22]. The pre-processing is done by 2 methods here: -

1) *Stemming*: The process of removing inflected words basically from stem words which are affixes, we remove the prefixes and suffixes to gain the root word. The term word stem is embraced from linguistics and utilized to relate to the base or root shape of a word. For illustration, words like 'programmer', 'programming', 'program' will be stemmed to 'program'. Typically, a lookup table is exercised to detect the word and its corresponding stem. Numerous search engines apply stemming for re-collecting documents that match user inquiries. Stemming is also utilized at the pre-processing stage for operations similar as emotion identification and textbook bracket. Stemming works on some languages substantially English and Spanish.

2) *Lemmatization*: Lemmatization involves converting all words to their corresponding root words, called "lemma". While stemming reduces all words to their root words via a lookup table. This means stemming can't differentiate which meaning of the word right is intended in the sentences "Please turn right at the coming light" and "She's invariably right". The stemmer would stem right to right in both rulings; the

lemmatizer would treat right other based upon its operation in the two expressions.

A lemmatizer also converts nonidentical word forms or inflections to a standard-issue shape. For illustration, it would convert lower too little, scratched to write, snoozed to sleep, etc. A lemmatizer works with further regulations of the language and contextual information than does a stemmer. It also relies on a wordbook to look up matching words. Because of that, it requires further processing authority and time than a stemmer to induce output. For these reasons, some NLP operations only exercise a stemmer and not a lemmatizer.

C. Model for Text Analysis

The sentiment analyzer categories the given text as sentiment and will give us the result based on positive, negative and neutral. Because of the better result we basically used three models for the sentiment analysis as: -

1) *Bag of Words*: A large feature vector known as a "bag of words" shows the frequency of various terms from a corpus in the provided text. The approach does not consider the word order at all. For many NLP applications, basic features like word and character n-grams are frequently used. The feature transformation is carried out by representing words and using n-grams. [23,24].

2) *Word Embeddings*: The semantic way to put out words is a word2vec approach [25]. They are represented as dimensional vector. In dimension vector we have syntactic and semantic words. We will implement distributed memory and distributed bag of words both [27]. Previously the same configuration was used as word embedding in which there was a Circle dataset [25]. The paragraph vector is also considered. Paragraph vector is a kind of extension of word embedding [26].

3) *TF-IDF*: TF-IDF stands for Term Frequency Inverse Document Frequency which is defined as calculation of way phrase in a sequence or corpus which is applicable to a textual content. It shows how important a particular word is with statistic and will increase proportionally to the wide variety of instances where that word is used. We used it as it is easy to calculate and computationally reasonably priced and is an easy place to begin for similarity calculations. It even offers much higher accuracy than BERT [28].

D. Performing Evaluation Process

Stock market prices to predict the future needs sequential data. The data should be mapped and work conjointly to work accordingly. We considered a few machine learning algorithms with different parameters. LSTM, ARIMA and SARIMA are used and implemented to compare and get better results.

1) *LSTM*: LSTM is basically a model of RNN which is first given by Hochreiter [29][30]. Recurrent neural networks (RNNs) in the NLP community are very popular these days. The LSTM cells are implemented in hidden layers[31]. These LSTM cells helps to remember the previous information and steps. That is why sequential data LSTM are the best. The normal layers and RNN had the problem of dependancy which

is for the long range. It also has a grading and vanishing problem [32], for that reason we choose LSTM [29]. A pseudo code would clearly explain how are we going to perform the methods and steps in a sequence.

Algorithm 1 LSTM

```

1: Begin
2: Initialize dataframe to news articles
3: while Not_Empty(Dataframe) do
4:   Dataframe  $\leftarrow$  Remove_Space
5:   Dataframe  $\leftarrow$  Remove_end_line_break
6:   Dataframe  $\leftarrow$  Remove_rows_without_publishdate
7:   Dataframe  $\leftarrow$  Remove_duplicate_titles
8:   Dataframe  $\leftarrow$  Sort_ascending
9:   Dataframe  $\leftarrow$  Remove_duplicate_dates
10:  Dataframe  $\leftarrow$  Remove_stop_words
11:  Dataframe  $\leftarrow$  Stemming
12:  Dataframe  $\leftarrow$  Lemmetization
13: end while
14: dataframe  $\leftarrow$  articles
15: Tokenization
16: Tf_idf  $\leftarrow$  TF_IDF
17: Bog  $\leftarrow$  Bag_of_words
18: Word_embedding  $\leftarrow$  word2vec
19: train_data  $\leftarrow$  80%_dataframe
20: test_data  $\leftarrow$  20%_dataframe
21: epoch  $\leftarrow$  200
22: Initialize LSTM Model by using sequential object
23: Train the model, monitor training loss;
24: Continue Until training loss at epoch  $n <$  training loss
25: at epoch  $n + 1 < \bullet <$  training loss at epoch  $n + 5$ , Or
    maximum epochs are reached.
26: Evaluate model on the test data.
27: Calculate RMSE.
28: Calculate minimum, maximum, average and standard deviation of RMSE.
29: End

```

2) *ARIMA*: ARIMA model is a linear model which is used for predicting. ARIMA model is break down into AR, I and MA. we've got used AR and MA in our sensible as Integrated (I) represents the distinction that is supposed to be applied to create the info stationary. The extreme values are handled as the outlier can be problematic for the for-cast as they lie outside the final trend which is therefore captured by the model. Following is the Pseudo code which is used for the ARIMA model with the orders.

a) *Autoregressive (AR)*: In an autoregression model, we have a tendency to basically predict the variable and even forecast it by the linear combination which should be there as the past value. The variable is the regression of itself. After this we get a tendency to target variable as a lagged value and get the output variable to predict the values for a longer term.

The best values with the normal way is to put $p=1$ and get the Partial Autocorrelation which performs plot. The ACF is mainly the correlation within the past and present values

Algorithm 2 ARIMA

```

1: Begin
2: Initialize dataframe to news articles
3: while Not_Empty(Dataframe) do
4:   Dataframe  $\leftarrow$  Remove_Space
5:   Dataframe  $\leftarrow$  Remove_end_line_break
6:   Dataframe  $\leftarrow$  Remove_rows_without_publishdate
7:   Dataframe  $\leftarrow$  Remove_duplicate_titles
8:   Dataframe  $\leftarrow$  Sort_ascending
9:   Dataframe  $\leftarrow$  Remove_duplicate_dates
10:  Dataframe  $\leftarrow$  Remove_stop_words
11:  Dataframe  $\leftarrow$  Stemming
12:  Dataframe  $\leftarrow$  Lemmetization
13: end while
14: dataframe  $\leftarrow$  articles
15: Tokenization
16: Tf_idf  $\leftarrow$  TF_IDF
17: Bog  $\leftarrow$  Bag_of_words
18: Word_embedding  $\leftarrow$  word2vec
19: Procedure ARIMA Model
20: aic  $\leftarrow$  inf
21: for  $p = 0$  to  $3$  do
22:   for  $d = 0$  to  $1c$  do
23:     for  $q = 0$  to  $3$  do
24:       model+fit(arima(p,d,q,allowdrift  $\leftarrow$ 
        True allowmean  $\leftarrow$  True),  $x$ )
25:       aic_curr  $\leftarrow$  compute_AIC(model)
26:       if aic_curr  $\leftarrow$  aic then
27:         model_opt  $\leftarrow$  model
28:         aic  $\leftarrow$  aic_curr
29:       End

```

of a constant variable. It perform together for the translativ impact that can be carried with the time aside by an on-the-spot impact. For instance, costs of oil a pair of days alone can influence costs one day alone and eventually, today. However, the costs of oil a couple of days alone may additionally influence these days that are measured. PACF measures on the other hand the past and current values for an instance. PACF can solely live the impact of costs of oil a pair of days ago on these days with no translativ impact.

b) *Moving Average(MA)*: The Moving Average term, depicted by the parameter 'q', refers to the quantity of lagged forecast errors wanted to predict future values. it's wont to cut back noise in a very model or sleek it out. The longer the moving average amount, the more smoothed out the noise would be. Since 'e' will solely be given once determining the model and changing it parameters too thus during this case we get a parameter named as 'e' which is unpredictable.

3) *SARIMA*: SARIMA is a model which is basically a seasonality ARIMA. When the ARIMA cannot sum up the data then the significance of seasonality is applied.

The part of these models are same like the ARIMA model as AR, I and MA which are stick around to SARIMA with the addition of Seasonality which changes it characteristics

and adds robustness.

There are uppercase and lowercase notations which we use in SARIMA. The uppercase are for seasonality function and lowercase as a part of model which is also used in ARIMA. The ACF and PACF plots helps the data to derive and check seasonality. By using these three algorithms, we are to compare which will show us the best result. The pseudo code has been given below for SARIMA.

Algorithm 3 SARIMA

```

1: Begin
2: Initialize dataframe to news articles
3: while Not_Empty(Dataframe) do
4:   Dataframe  $\leftarrow$  Remove_Space
5:   Dataframe  $\leftarrow$  Remove_end_line_break
6:   Dataframe  $\leftarrow$  Remove_rows_without_publishdate
7:   Dataframe  $\leftarrow$  Remove_duplicate_titles
8:   Dataframe  $\leftarrow$  Sort_ascending
9:   Dataframe  $\leftarrow$  Remove_duplicate_dates
10:  Dataframe  $\leftarrow$  Remove_stop_words
11:  Dataframe  $\leftarrow$  Stemming
12:  Dataframe  $\leftarrow$  Lemmetization
13: end while
14: dataframe  $\leftarrow$  articles
15: Tokenization
16: Tf_idf  $\leftarrow$  TF_IDF
17: Bog  $\leftarrow$  Bag_of_words
18: Word_embedding  $\leftarrow$  word2vec
19: Procedure SARIMA Model
20: bic  $\leftarrow$  inf
21: for  $(p, d, q)(P, D, Q, s)$  do
22:   Model  $\leftarrow$  FitSARIMA( $p, d, q)(P, D, Q, s)$ 
23:   Selectbest( $p, d, q)(P, D, Q, s)$ 
24: end for
25: Evaluate model on the test data.
26: Calculate RMSE.
27: End =0

```

IV. RESULTS AND DISCUSSION

The main thinking of doing our research is basically to predict better accuracy and result for analyzing the Stock market predictions. We have taken and compared the FTSE 100 index with the main companies listed by the primary market capitalization in EUROPE. We have considered as 5 Top companies named as AstraZeneca, Unilever, HBSC, BP, GALXO. The adjusted close price will look like the one stated below. (Fig.1).

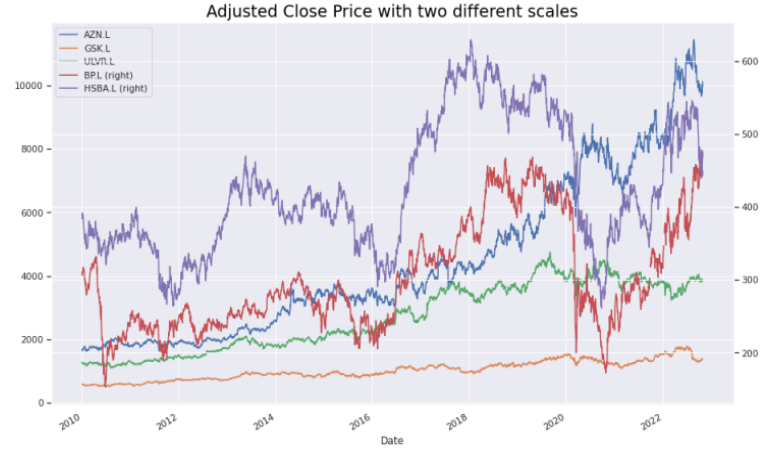


Fig. 1. Adjusted Close Price

For our research we started with ARIMA in which we had ADF test and ACF. Even we did the differencing as parameter 'd' to find the appropriate order in time series. Then Auto Regressive with Moving Average is also performed to make the ARIMA model. After choosing the best grid for model we do the prediction which is very close to the actual values. We can forecast the next 60 days and visualize the data points for Adjusted Close prices (Fig.2).



Fig. 2. Plot actual vs Predicted Result for ARIMA

As for SARIMA we will build, train and test a SARIMA model for predicting AstraZeneca Adjusted Close price patterns. SARIMA is bit same as ARIMA, so we followed the same steps and worked with ACF and PACF. The Augmented Dickey-Fuller test and differencing was done. A SARIMAX implementation is so-called because the X addition means that it also supports exogenous variables. This is an optional argument, rather than a parameter, for parallel time series variety made available as a weighted input to the model. After the modeling as we select the best grid, we do the training and testing. The RMSE value is 154.733, not as good as LSTM. The train, actual, forecast and fit to training data is shown through plot graph as described (Fig.3).



Fig. 3. Forecast Vs Actual Test value for SARIMA

Stock prices do not have a particular trend or seasonality, being highly dependent on market forces which cause prices to fluctuate. As a result, forecasting techniques such as ARIMA and SARIMA do not yield good results for this problem, but LSTM shows a much better result than ARIMA and SARIMA. When we consider short term forecasting results we consider ARIMA over LSTM but when we consider long term predictions we always choose LSTM. What we want is to make our short term forecasting more better and accurate so ARIMA would be preferred and even it turns out to give best RMSE score too. LSTM has the ability to store information over a period and is useful when dealing with time series data. So, five years of AstraZeneca historical stock data will be used to predict Adjusted Close prices by building a multi-layer LSTM Recurrent Neural Network model.

We split the trained data into X_{train} and Y_{train} and consider about 200 epochs. Calculations of RMSE are done using the actual value as lower values of RMSE indicate a better fit. As we got 78.81% accurate results for LSTM. Accuracy could be improved by trying different numbers of layers, epochs and batch sizes. The results would be like (Fig.4) in which we have training, actual and predicted graph.



Fig. 4. Plot actual vs predicted values for LSTM

The LSTM has achieved the lowest RMSE and the most accurate predictions. This is just a basic model, but various parameters can be tuned, more layers added with dropout for regularization and an increased number of epochs for better convergence.

V. CONCLUSION

As a result of the stock market's close relationship to a country's economic growth, the market draws sizeable investments from investors, and it issues securities in the public interest, forecasting the movement of the stock prices and the market becomes essential in order to prevent significant losses and make wise decisions. We offered a comparative analysis of numerous algorithms for predicting the prices of various stocks in this work. The system for stock price prediction that effectively combined sentimental analysis for fundamental analysis with long short-term memory for technical analysis gave good accuracy. Based on the focused keyword, a sentiment analysis was conducted. The study also incorporates a number of additional methodologies, such as time series analysis, and graph-based algorithms, and compares the outcomes of these algorithms. The user who has no prior knowledge of the stock market would benefit from this system. Users of all trading skill levels are able to utilize this approach to forecast future stock prices.

Future versions of the system would make inter-day forecasting possible and enhance sentimental analysis to produce a sarcasm-free impact factor for fundamental analysis. Future research can focus on combining information from stock sentiment analysis with quantitative data corresponding to historical stock values to forecast stock prices. More effective stock recommendation algorithms could be developed by combining the two pieces of information. Deep learning-based procedures may be used more often to provide feature extraction techniques that are more efficient. Although Graph knowledge techniques are promising and can be utilized to develop stock prediction engines, research should focus on the complexity and gradient of large-scale graphs.

VI. ACKNOWLEDGEMENTS

The authors are thankful for the supervision, support and guidance from Dr. Jinan Fiaidhi, PhD, ISP, IEEE Senior Member, PEng (Full Professor) the Department of Graduate studies, Lakehead University, Thunderbay, Ontario, Canada

REFERENCES

- [1] Gurjar, M., P. Naik, G. Mujumdar, and T. Vaidya. 2018. "Stock Market Prediction Using ANN." *International Research Journal of Engineering and Technology (IRJET)*, no. 03. <https://doi.org/10.15623/ijret.2014.0304132>
- [2] Qian, B., and K. Rasheed. 2007. "Stock Market Prediction with Multiple Classifiers." *Applied Intelligence* 26: 25–33. <https://doi.org/10.1007/s10489-006-0001-7>
- [3] Stambaugh, Robert F., Jianfeng Yu, and Yu Yuan. 2012. "The Short of It: Investor Sentiment and Anomalies." *Journal of Financial Economics* 104 (2): 288–302. <https://doi.org/10.1016/j.jfineco.2011.12.001>
- [4] Ohana, Jean-Jacques, Steve Ohana, Eric Benhamou, David Saltiel, and Beatrice Guez. 2021. "Explainable AI Models of Stock Crashes: A Machine-Learning Explanation of the Covid March 2020 Equity Melt-down." *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3809308>

- [5] Mohan, Saloni, Sahitya Mullapudi, Sudheer Sammeta, Parag Vijayvergia, and David C. Anastasiu. 2019. "Stock Price Prediction Using News Sentiment Analysis." IEEE Xplore. April 1, 2019. <https://doi.org/10.1109/BigDataService.2019.00035>.
- [6] Pavithya, M. B. D., G. S. D. Perera, S. L. Munasinghe, and S. N. Karunarathna. 2021. "Quantitative Analysis and Sentiment Analysis for Stock Price Forecast: The Case of Colombo Stock Exchange." 2021 10th International Conference on Information and Automation for Sustainability (ICIAFS), August. <https://doi.org/10.1109/iciafs52090.2021.9606034>.
- [7] Chen, Ray and Marius Lazer. "Analysis of Twitter Feeds for the Prediction of Stock Market Movement." (2011).
- [8] Zhang, Linhao. 2013. "Sentiment Analysis on Twitter with Stock Price and Significant Keyword Correlation." Repositories.lib.utexas.edu. 2013. <http://hdl.handle.net/2152/20057>
- [9] Hassan MR, Nath B. Stock market forecasting using Hidden Markov Model: a new approach. In: Proceedings—5th international conference on intelligent systems design and applications 2005, ISDA'05. 2005. pp. 192–6. <https://doi.org/10.1109/ISDA.2005.85>.
- [10] Cervelló-Royo, Roberto, Francisco Guijarro, and Karolina Michniuk. 2015. "Stock Market Trading Rule Based on Pattern Recognition and Technical Analysis: Forecasting the DJIA Index with Intraday Data." Expert Systems with Applications 42 (14): 5963–75. D. Li and J. Qian, "Text sentiment analysis based on long short-term memory," 2016 First IEEE International Conference on Computer Communication and the Internet (ICCCI), Wuhan, China, 2016, pp. 471–475, doi: 10.1109/CCCI.2016.7778967.
- [11] Arévalo, Andrés, Jaime Niño, German Hernández, and Javier Sandoval. 2016. "High-Frequency Trading Strategy Based on Deep Neural Networks." In Intelligent Computing Methodologies, 424–36. https://doi.org/10.1007/978-3-319-42297-8_40.
- [12] Lin, Bo-Sheng, Wei-Tao Chu, and Chuin-Mu Wang. 2018. "Application of Stock Analysis Using Deep Learning." In 2018 7th International Congress on Advanced Applied Informatics (IIAI-AAI). <https://doi.org/10.1109/iiat-aa.2018.00128>, India, 2020, pp. 1–6, doi: 10.1109/ICCSEA49143.2020.9132928
- [13] Y Bakshi, Rushlene Kaur, Navneet Kaur, Ravneet Kaur and Gurpreet Kaur. "Opinion mining and sentiment analysis." 2016 3rd International Conference on Computing for Sustainable Global Development (INDIACom) (2016): 452–455.
- [14] Musto, Cataldo, Giovanni Semeraro and Marco Polignano. "A Comparison of Lexicon-based Approaches for Sentiment Analysis of Microblog Posts." DART@AI*IA (2014).
- [15] Oliveira, Nuno, Paulo Cortez, and Nelson Areal. 2014. "Automatic Creation of Stock Market Lexicons for Sentiment Analysis Using StockTwits Data." Proceedings of the 18th International Database Engineering Applications Symposium on - IDEAS '14. <https://doi.org/10.1145/2628194.2628235>.
- [16] Schumaker Chen (2009) Schumaker RP, Chen H. Textual analysis of stock market prediction using breaking financial news: the AZFin text system. ACM Transactions on Information Systems. 2009;27(2):1–19
- [17] Sirignano J, Cont R. Universal features of price formation in financial markets: perspectives from deep learning. Ssrn. 2018. <https://doi.org/10.2139/ssrn.3141294>.
- [18] Ding X, Zhang Y, Liu T, Duan J. Using structured events to predict stock price movement: An empirical investigation. Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP).2014.
- [19] <https://ca.finance.yahoo.com/>
- [20] Mohan, S., Mullapudi, S., Sammeta, S., Vijayvergia, P., Anastasiu, D. C. (2019). Stock Price Prediction Using News Sentiment Analysis. 2019 IEEE Fifth International Conference on Big Data Computing Service and Applications (BigDataService).
- [21] <https://www.investing.com/>
- [22] B. Pang, L. Lee, and S. Vaithyanathan, "Thumbs up? Sentiment classification using machine learning techniques," CoRR, vol. cs.CL/0205070, 2002. R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.
- [23] Y.Mao, W. Wei, B. Wang, and B. Liu, "Correlating sp 500 stocks with twitter data," in Proceedings of the First ACM International Workshop on Hot Topics on Interdisciplinary Social Networks Research, 2012, pp. 69–72.
- [24] J. Si, A. Mukherjee, B. Liu, S. J. Pan, Q. Li, and H. Li, "Exploiting social relations and sentiment for stock prediction." in EMNLP, vol. 14, 2014, pp. 1139–1145.
- [25] T. Mikolov, I. Sutskever, K. Chen, G. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," CoRR, vol. abs/1310.4546, 2013.
- [26] Q. Le and T. Mikolov, "Distributed representations of sentences and documents," in Proceedings of the 31st International Conference on Machine Learning (ICML-14), 2014.
- [27] R. Rehurek and P. Sojka, "Software Framework for Topic Modelling with Large Corpora," in Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks, 2010, pp. 45–50.
- [28] Anirudha Simha, "Understanding TF-IDF for Machine Learning", 06-Oct2021. Available: <https://www.capitalone.com/tech/machinelearning/understanding-tf-idf>.
- [29] D. Li and J. Qian, "Text sentiment analysis based on long short-term memory," 2016 First IEEE International Conference on Computer Communication and the Internet (ICCCI), Wuhan, China, 2016, pp. 471–475, doi: 10.1109/CCCI.2016.7778967.
- [30] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, pp. 1735–1780, 1997.
- [31] A. Sarkar, A. K. Sahoo, S. Sah and C. Pradhan, "LSTMSA: A Novel Approach for Stock Market Prediction Using LSTM and Sentiment Analysis," 2020 International Conference on Computer Science, Engineering and Applications (ICCSEA), Gunupur, India, 2020, pp. 1–6, doi: 10.1109/ICCSEA49143.2020.9132928
- [32] Y. Bengio, P. Simard, and P. Frasconi, "Learning long-term dependencies with gradient descent is difficult," Trans. Neur. Netw., vol. 5, no. 2, pp. 157–166, Mar. 1994.
- [33] Available: <https://neptune.ai/blog/arima-sarima-real-world-time-series-forecasting-guide>